

# When the World Changes but the Score Does Not: A Counterfactual Audit of Computer-Use Agents

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## Abstract

Computer-use agents are typically evaluated by whether they complete a task under a fixed rubric. Task completion does not establish that the score *identifies* which environment state produced the success. We audit this property with paired counterfactual episodes: a controlled intervention changes a pre-registered determining set  $D$  while the instruction, interface, rubric, and harness stay fixed. We separately code *tracking* (does the final answer follow  $D$ ?) and *score invariance* (does the judge move when the gold moves?).

Under a frozen Stage-4 protocol on MyPCBench, three computer-use systems (Claude Opus 4.6, GPT-5.5, Qwen3.5-35B-A3B) all track a loyalty-tier intervention on `retrieval-f001` while the judge stays at 100. Claude and GPT also track an aggregation intervention ( $\approx \$4872 \rightarrow \$400$ ) with unchanged scores. A Qwen3.5-9B size ablation tracks both tasks; an exploratory Qwen3.8-Flash run does too. Score invariance does *not* hold on every completed Qwen pair. Incomplete trajectories are coded as technical failures, not as “does not track.” On Claude `preference-inference-f018`, the judge stays at 100 while tracking of the joint  $D$  is only partial—the one current cell in which success conceals incomplete tracking.

The claim is not that agents ignore personal data, nor that invariant scores are always wrong. It is that a scalar task score is an insufficient identifier of counterfactual world-state. For CUA development this is operational, not only diagnostic: a stable leaderboard number is not evidence that the agent is latched onto a particular personal record, and a high score can conceal two incompatible worlds—or an incomplete read of a joint determining set. Paired gold-moving checks belong beside completion scores; conventional CUA leaderboards do not perform them.

## 1 Introduction

Computer-use agents (CUAs) act on screenshots, click and type, and are scored on whether they finish the task [Jang et al., 2026, Xie et al., 2024]. That evaluation answers a capability question: *did the trajectory satisfy the rubric?* It does not answer a reliability question:

*When a task-relevant state in the world is counterfactually changed, does the agent’s reported outcome change with it—and would the score have told us?*

These are different questions. Consider a loyalty dashboard. An intervention changes Gold Voyager / 38,450 miles into Silver Voyager / 8,620. If the agent then reports Silver / 8,620, it tracked the state. If the judge remains 100, the score did not identify which world produced the

success. Conversely, if the agent still reports Gold while scoring 100, the score conceals a tracking failure.

We therefore treat CUA evaluation as a *counterfactual audit*, not as another benchmark sweep. For each frozen task we run a baseline environment  $E^0$  and an intervened environment  $E^1 = I(E^0)$ , with  $I$  locked before confirmatory runs. The same instruction, rubric, and agent interface are applied to both. From each completed pair we read two quantities that the locked protocol already distinguished:

- **Tracking** (behavioral DV): does the final answer follow the determining set  $D$  of the intervention?
- **Score invariance** (evaluator diagnostic): is  $|S^1 - S^0|$  small relative to the gold change, given that both episodes completed and a guest probe shows that gold moved?

The object of study is interactive computer-use agents under controlled SQL interventions.

### Contributions.

1. A counterfactual audit that separates tracking from task scoring, with explicit coding of technical failure vs. behavioral evidence.
2. A frozen intervention protocol (eligibility, seed, SQL, DVs) on MyPCBench [Jang et al., 2026], using real  $1280 \times 800$  screenshots and guest-side probes that gold moved.
3. Cross-agent evidence on two clean tasks, a within-family 9B ablation, an exploratory Flash check, and one Claude cell in which score invariance coexists with incomplete tracking.

**What we do not claim.** We do not claim that agents fail to use personal data. On the clean pairs they often *do* track. We do not claim model-generalizability from two or three systems. We do not promote held-out or confounded task IDs after seeing outcomes. The Phase-A memo locked these restrictions before the GPT and Qwen cells.

## 2 Related work

**Computer-use evaluation.** Desktop and web agents are scored in interactive environments [Xie et al., 2024, Zhou et al., 2024, Koh et al., 2024]. MyPCBench [Jang et al., 2026] additionally seeds a coherent persona across 17 logged-in apps and 184 tasks, and reports rubric scores under a computer+bash tool surface. Those leaderboards measure completion. They do not intervene on the persona state that the rubric treats as gold and then ask whether the score still identifies the world.

**Reliability beyond accuracy.** A score can be high for reasons other than the intended mechanism. If the object of interest is state-sensitive behavior, a scalar that is invariant to a gold-moving intervention is not a sufficient statistic for that object.

**Counterfactuals as identification.** The design is a paired intervention [Pearl, 2009]: same task, same agent, one pre-registered state change. Unrelated-task comparisons cannot separate agent behavior from task difficulty.

### 3 Counterfactual audit

#### 3.1 Two DVs, not one

For frozen task  $t$ , let  $E_t^0$  be the seeded guest and  $E_t^1 = I_t(E_t^0)$  the guest after a locked SQL patch on determining set  $D_t$ . The same agent  $A$ , instruction  $T$ , screenshot size, and judge  $J$  yield trajectories  $\tau^0 = A(T, E_t^0)$  and  $\tau^1 = A(T, E_t^1)$ . A guest probe after  $I_t$  must show that gold moved; otherwise the pair is not an intervention.

From each completed  $\tau$  we extract  $G$  (the task-relevant quantities in the *final answer*) and  $S = J(\tau)$  (rubric score in  $\{0, \dots, 100\}$ ).

**P2 — tracking (primary behavioral DV).** Code track = 1 iff  $G^1$  matches  $D_t$  on  $E_t^1$  and  $G^0$  matches  $D_t$  on  $E_t^0$ , using the pre-registered answer fields (tier and miles; combined refund; asserted GME shares and YES bet; etc.). Do not code tracking from  $S$ .

**P1 — score invariance (evaluator diagnostic).** If both episodes reach DONE, the probe shows gold moved, and  $|S^1 - S^0|$  is small relative to the gold change, the judge did not detect the intervention. This is *not* a claim that the agent is reliable. It is a claim about  $J$ .

**Why invariance can be expected.** Eligibility required channel-invariant rubrics that do not pin a dollar/tier gold in the instruction. Under that screen, a judge that scores “did the agent use the apps and report a loyalty tier?” can award 100 to both Gold and Silver. Score invariance is then a property we *built into* the sample, not a surprise about Claude’s internals. It remains scientifically useful: it shows that published-style success is compatible with two incompatible worlds.

#### Two patterns that matter.

##### Type A.

Track and score-invariant. The agent read the new world; the score cannot tell Gold from Silver. This is an evaluation gap.

##### Type B.

Incomplete tracking and score-invariant. The agent did not follow the full  $D$ ; the score still looks like success. This is a reliability gap the leaderboard hides.

Type A is currently the replicated pattern. Type B is currently one Claude cell (f018). They must not be averaged into one “agents are unreliable” slogan.

#### 3.2 Coding incomplete episodes

A GUI abort, step-limit without DONE, missing tool XML, or screenshot-decode crash is P3: `technical_failure` / `failure`. It is not track = 0 and not a refutation of P1. Empty trajectories and host crashes are infrastructure, not DVs.

### 3.3 Frozen protocol

Selection was pre-registered on eligibility A–E (channel-invariant; mapped SQLite; `cua_required` false; no LibreOffice/GUI artifact; instruction does not pin a dollar gold) and a seed 20260826 sample,  $k \leq 3$  per evidence type:

$$184 \rightarrow 10 \text{ eligible} \rightarrow 8 \text{ confirmatory sample.}$$

Four sample members are confounded on this seed and stay in the sample un-run for attribution: `retrieval-f029`, `retrieval-f030`, `aggregation-f018`, `preference_inference-f004`. Two identifiable reserves (`retrieval-f003`, `retrieval-f016`) were not promoted after seeing outcomes.

Runnable Stage-4 IDs: `retrieval-f001`, `aggregation-f003`, `preference_inference-f018`, `counterfactual-f004`. SQL patches are locked in `cf/stage4_locked.json`. We do not invent a different patch after seeing an answer.

## 4 Experimental setup

**Environment.** Official MyPCBench QEMU guest, persona `michael.scott@dundermifflin.com` (canonical identity), screenshots  $1280 \times 800$ . Interventions are SQL on guest SQLite; the agent is not told that a patch occurred. Judge: Anthropic `per_step` when available (auxiliary).

### Agents.

- Claude: `claude_cuabash` / `claude-opus-4-6` (confirmatory Stage 4).
- GPT: `openai_cuabash` / `gpt-5.5` (exploratory transfer; same SQL).
- Qwen paper CUA: `qwen_cuabash` / `qwen/qwen3.5-35b-a3b` via a hosted OpenAI-compatible endpoint (OpenRouter). Local HPC 27B+vLLM was abandoned after infrastructure failure and is *not* a result.
- Qwen3.5-9B: size ablation under the same XML CUA interface, `f001+f003` only.
- Qwen3.8-Flash: exploratory cross-version check, same two tasks. It does not replace 35B-A3B as the paper Qwen id.

## 5 Results

### 5.1 How to read the tables

Table 1 is the scan table: only DONE+DONE pairs, with  $\Delta S = S^1 - S^0$  in its own column. Primary Type A rows have  $\Delta S = 0$ . The two Qwen size/exploratory `f003` rows are the completed pairs where the judge *does* move. Table 2 keeps incomplete cells in view; they are not in any invariance denominator. Writer scripts that mark a crashed cell as `ok` are overridden by the trajectory (Qwen 35B `f018` CF is `PREDICT_CRASH`).

Same task / same rubric / same harness

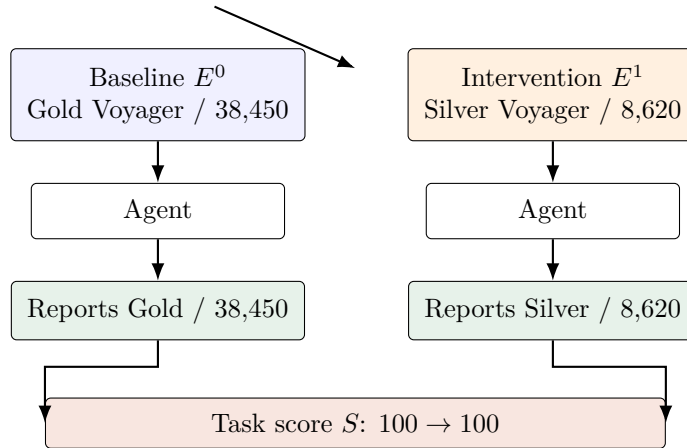


Figure 1: **Type A audit (schematic of retrieval-f001)**. The world state changes by an order of magnitude; the agent tracks it; the conventional score need not move. Caption, not architecture diagram: world changed; behavior tracked; score was not an identifier.

## 5.2 Cross-agent Type A on f001

retrieval-f001 is the cleanest cell.  $I$  updates `loyalty.{status, miles, miles_ytd}` from Gold Voyager / 38,450 to Silver Voyager / 8,620. Claude, GPT, and Qwen3.5-35B-A3B each complete both episodes, report the new tier and miles, and keep  $S = 100$ . Figure 1 is this cell.

This is the strongest cross-agent result we have: three independent CUA stacks, one locked SQL, Type A. It is still one task and one evidence type (`point_field`).

**Primary  $k/n$ .** Across the frozen primary agents, there are five valid DONE+DONE pairs on f001+f003 (35B-A3B f003 excluded). All five track  $D$  and all five are score-invariant: 5/5 (Clopper–Pearson 95% CI [0.48, 1.00]). We do not interpret the interval as a significance test. 9B f003 (50 → 80) and Flash f003 (80 → 100) are tracking-valid and *score-sensitive*; they are not in this denominator.

## 5.3 Aggregation: f003

$I$  rewrites filed 2023/2024 refunds so the combined gold moves 4871.70 → 400.00 ( $n_{\text{filed}} = 2$  held fixed; 2025 in-progress untouched). Claude tracks with 80 → 80. GPT tracks with 50 → 50 (the absolute score is lower—the GPT answer missed year-count rubric items—but P1 still holds). Qwen3.5-35B-A3B completes the baseline ( $\approx \$4872$ ) and does not complete the CF (attempt 1: 80-step scroll loop; later retry: FAIL / missing executable tool call). That is missing evidence, not “Qwen does not track \$400.”

Qwen3.5-9B tracks both sides.  $S$  moves 50 → 80, so this pair supports tracking and *rejects* a blanket invariance claim. Qwen3.8-Flash likewise tracks;  $S$  moves 80 → 100. We report those deltas rather than folding them into “the same qualitative pattern.”

| Agent                                                             | Task | $S^0$ | $S^1$ | $\Delta S$ | Track   | Pattern               |
|-------------------------------------------------------------------|------|-------|-------|------------|---------|-----------------------|
| <i>Primary Type A — invariance 5/5, <math>\Delta S = 0</math></i> |      |       |       |            |         |                       |
| Claude                                                            | f001 | 100   | 100   | 0          | yes     | Type A                |
| Claude                                                            | f003 | 80    | 80    | 0          | yes     | Type A                |
| GPT-5.5                                                           | f001 | 100   | 100   | 0          | yes     | Type A                |
| GPT-5.5                                                           | f003 | 50    | 50    | 0          | yes     | Type A                |
| Qwen 35B-A3B                                                      | f001 | 100   | 100   | 0          | yes     | Type A                |
| <i>Type B (one cell; not in the 5/5 rate)</i>                     |      |       |       |            |         |                       |
| Claude                                                            | f018 | 100   | 100   | 0          | partial | Type B                |
| <i>Size ablation / exploratory — not pooled into 5/5</i>          |      |       |       |            |         |                       |
| Qwen 9B                                                           | f001 | 80    | 80    | 0          | yes     | Type A                |
| Qwen 9B                                                           | f003 | 50    | 80    | +30        | yes     | tracks; not invariant |
| Qwen Flash                                                        | f001 | 100   | 100   | 0          | yes     | Type A                |
| Qwen Flash                                                        | f003 | 80    | 100   | +20        | yes     | tracks; not invariant |

Table 1: Valid DONE+DONE pairs. Gold moved on every row (f001: Gold/38450  $\rightarrow$  Silver/8620; f003:  $\approx$  \$4872  $\rightarrow$  \$400; f018: GME 85  $\rightarrow$  0 and OM YES 200 active $\rightarrow$ settled).  $S$  is the auxiliary rubric score. Primary invariance is 5/5 with  $\Delta S = 0$  (35B f003 has no valid CF). 9B and Flash f003 show that  $\Delta S = 0$  is not automatic once tracking succeeds.

| Agent        | Task | Why excluded from $\Delta S$ | Last status                                |
|--------------|------|------------------------------|--------------------------------------------|
| Claude       | f004 | no baseline DONE             | $S$ : 0 $\rightarrow$ 87; out of inference |
| GPT-5.5      | f018 | no DONE either cell          | not “does not track”                       |
| GPT-5.5      | f004 | 1099 still cited after patch | 87 $\rightarrow$ 87; not attributed        |
| Qwen 35B-A3B | f003 | CF not DONE (3 attempts)     | missing, not negative                      |
| Qwen 35B-A3B | f018 | CF screenshot decode crash   | technical failure                          |
| Qwen 35B-A3B | f004 | step limit, both sides       | failure                                    |

Table 2: Cells that do not enter tracking or invariance denominators. Execution failure is not coded as track = 0.

## 5.4 Type B: Claude f018

The locked  $D$  is a *conjunction across two apps*: BatBucks GME shares 85  $\rightarrow$  0 and OddsMarket YES 200/active  $\rightarrow$  0/settled. The instruction asks for a GameStop-conviction rebalance (cost basis only). Claude’s CF answer tracks the stock conjunct (shares = 0) and still asserts an active YES 200 bet.  $S$  stays 100  $\rightarrow$  100. GPT and Qwen 35B did not produce a valid CF DONE; we do not assign Type B to them.

This is not “the agent ignored personal data.” It read BatBucks. The gap is that  $J$  can treat a coherent GME-conviction narrative as task success without requiring both conjuncts of  $D$  to be represented in the final answer. Two structural features make that possible. First,  $D$  is multi-source: zeroing one database does not automatically surface the other in the same screenshot. Second, the rubric is advice-shaped rather than a pinned field check, so reporting “lean into GameStop” can still pass after the YES position is gone. Type A tasks (f001, f003) ask for a quantity that lives in one place; Type B here is what a joint  $D$  looks like when  $J$  scores the story, not each conjunct.

We therefore treat f018 as a mechanism sketch, not a replicated rate: a high score can conceal *which subset* of  $D$  produced the answer. Replicating Type B needs another completed pair on a

joint determining set—not another model on f001.

## 5.5 What we refuse to convert into DVs

Qwen 35B f004: both cells hit the step limit. Claude f004: no baseline DONE. GPT f018: no actions / no DONE. Qwen 35B f018 CF: screenshot decode crash. A hosted 35B interface layer can break XML tool emission; that is harness/model-interface, not a tracking estimate.

# 6 Discussion

## 6.1 The finding, stated narrowly

On completed gold-moving pairs, agents *can* track  $D$  while  $S$  stays put (Type A), and—in one Claude cell— $S$  can stay put while tracking of joint  $D$  is incomplete (Type B). A leaderboard number does not tell a reader which of these occurred, nor which world the agent saw. The 5/5 primary count is a census of valid primary pairs, not an estimate of Type A over 184 tasks. 9B/Flash f003 already show that tracking does not imply  $\Delta S = 0$ .

## 6.2 Score invariance is not automatically a bug

If  $I$  changed a variable the rubric ignores,  $S^0 \approx S^1$  is correct scoring. Our eligibility rule selected tasks whose instructions do not pin the personal quantity, so Type A is partly a statement about  $J$  under that screen. The operational implication is still sharp: *do not ship, rank, or select a CUA on a stable high score as if that score identified a particular personal record*. A paired gold-moving probe—same instruction, locked  $I$ , guest check that gold moved, tracking coded from the answer—is the test that the leaderboard omits. Type B adds a second operational warning: even when the agent used *some* personal evidence,  $J$  may not tell you whether it used the whole determining set.

## 6.3 Cross-agent and scale

Three families on f001 reduce the chance that Type A is a Claude artifact. 9B tracking both tasks reduces the chance that only the 35B-A3B checkpoint can see the dashboard. Neither fact licenses a 184-task sweep or a claim that size does not matter in general.

## 6.4 What would actually scale the claim

Paired-episode coding is already done: five primary pairs, all Type A; execution failures kept out of the denominator. A sixth CUA family would not change that. If we scale, the target is more *independent* gold-moving tasks of the same design—to test whether Type A is systematic or an artifact of f001/f003—plus item-level judge coding (whether  $J$  ever mentions the manipulated numeral). That still cannot manufacture Type B replication we do not have.

# 7 Limitations

- **N / generalizability.** This is an exhibited phenomenon on a frozen sample, not a population rate. Cross-agent Type A is strongest on one task (f001). Reviewers should treat 5/5 as a

complete count of valid primary pairs, not as evidence that Type A holds for CUA tasks in general.

- **Type B** is one Claude cell on a joint  $D$ . GPT/Qwen CF episodes on the same task did not complete, so the mechanism (conjunction + advice-shaped rubric) is not yet cross-agent.
- **Score movement** on 9B/Flash f003 already shows that P1 is not universal even inside the Qwen family.
- **Hosted Qwen 35B** is not the abandoned local vLLM 27B stack. Tool-call failures may mix model and endpoint.
- **We evaluate agent systems** (`claude_cuabash`, `openai_cuabash`, `qwen_cuabash`), not isolated base models.

## 8 Conclusion

We audited computer-use agents by changing the world and holding the task fixed. Conventional scores can remain compatible with two incompatible persona states; agents may track the change, track it only in part, or fail to finish the episode. Those three outcomes are not interchangeable, and a scalar rubric score does not name which one occurred. Counterfactual evaluation is a small complementary test: intervene on  $D$ , require a guest probe that gold moved, code tracking from the answer, and refuse to treat crashes as psychology. A completion score is not a substitute for that check.

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